AN EXPLICIT CLOSED FORM FOR THE FOUR-DIVISOR E_2 LAMBERT ANNIHILATOR AT SQUAREFREE COMPOSITE LEVEL: A NOTE ON AUTONOMOUS REDISCOVERY

JOUNI HELMINEN

Abstract. We give an explicit closed-form parameterization of the unique (up to scale) linear relation among the four Lambert "conjugate-pair blocks"

$$T_d(\alpha,\beta) := d\,\alpha\,L(d\alpha) + \frac{\beta}{d}\,L\!\left(\frac{\beta}{d}\right), \qquad d \mid N, \qquad L(x) := \sum_{n\geq 1} \frac{n}{e^{2nx}-1},$$

on the hyperbola $\alpha\beta = \pi^2$ at squarefree composite level $N = pq$ with distinct primes $p < q$. With $g := \gcd(pq-1,\, q-p)$, $\delta := (q-p)/g$, and $m := (pq-1)/g$, the integer-lowest-height coefficient vector is

$$(c_1, c_p, c_q, c_{pq}) = (-\delta,\ m,\ -m,\ \delta).$$

The derivation requires only the Hardy–Ramanujan–Berndt base identity (1916) plus five lines of Gaussian elimination on a 3×4 rational system. The underlying structural fact—that exactly one linear relation must exist—is a consequence of the classical dimension formula $\dim \mathcal{E}_2(\Gamma_0(pq)) = 3$ for the Eisenstein subspace of weight-2 modular forms at squarefree composite level. We verify the formula numerically across sixteen instances to relative error $\leq 10^{-100}$ at `mpmath` precision 200 digits, map structural extensions (the formula is unique at $N = pq$, non-unique at $N = pqr$ and $N = p^2q$, and does not extend to $N = p^2$), and offer the note as both a useful explicit parameterization and as an instance of autonomous rediscovery of classical material by an LLM-driven symbolic-search engine.

1. Introduction

In their 1916 paper *Asymptotic formulae in combinatory analysis* [6, eq. (1.55)], Hardy and Ramanujan recorded the following identity, equivalent to the modular transformation law for the weight-2 Eisenstein series E_2: for $\alpha, \beta > 0$ with $\alpha\beta = \pi^2$,

$$\alpha\,L(\alpha) + \beta\,L(\beta) = \frac{\alpha+\beta}{24} - \frac{1}{4}, \tag{1}$$

where $L(x) := \sum_{n\geq 1} n/(e^{2nx}-1)$ is the standard Lambert series. Berndt's commentary in *Ramanujan's Notebooks Part II* [2, Ch. 14, Entry 27(iii)] traces this identity to Ramanujan's second notebook and connects it directly to the E_2 inversion law

$$E_2(-1/\tau) = \tau^2\,E_2(\tau) + \frac{6\tau}{\pi i}, \tag{2}$$

via the substitution $\tau = i\alpha/\pi$ and the standard q-expansion $E_2(\tau) = 1 - 24\sum_{n\geq 1} n\,q^n/(1-q^n)$ with $q = e^{2\pi i\tau}$.

A natural question is: given a level N, what is the dimension of the linear-relation space spanned by the conjugate-pair blocks

$$T_d(\alpha,\beta) := d\,\alpha\,L(d\alpha) + \frac{\beta}{d}\,L\!\left(\frac{\beta}{d}\right) \tag{3}$$

Date: May 25, 2026.

2020 *Mathematics Subject Classification.* Primary 11F11, 11F20; Secondary 11M36, 68T20.

Key words and phrases. Lambert series; Eisenstein series; quasi-modular form E_2; level $\Gamma_0(N)$; Hardy–Ramanujan–Berndt identity; autonomous mathematical rediscovery.

indexed by $d \mid N$? For $N = 1$ there is one block and eq. (1) determines its value but gives no annihilator. For $N = p$ prime, there are two blocks $\{T_1, T_p\}$ but eq. (1) applied at the two conjugate pairs (α, β) and $(p\alpha, \beta/p)$ produces no non-trivial linear relation (both blocks are affine in (α, β) with linearly independent coefficient vectors).

The first non-trivial case is therefore $N = pq$ with $p \neq q$ both prime: four blocks indexed by $d \in \{1, p, q, pq\}$. The main result of this note is that this case admits exactly one non-trivial linear relation, with explicit closed-form coefficients.

Theorem 1.1. *Let $p < q$ be distinct primes. With $g := \gcd(pq - 1,\, q - p)$, $\delta := (q - p)/g$, and $m := (pq - 1)/g$, the unique (up to scale) integer-lowest-height annihilator of $\{T_1, T_p, T_q, T_{pq}\}$ on the hyperbola $\alpha\beta = \pi^2$ is*

$$-\delta\, T_1 + m\, T_p - m\, T_q + \delta\, T_{pq} = 0. \tag{4}$$

The remainder of this note is an exposition of eq. (4): we derive it in five lines (section 2), connect it to the classical E_2 Eisenstein-subspace dimension formula (section 3), tabulate sixteen numerically verified instances (section 4), map its structural extensions to richer divisor sets (section 5), and close with a provenance note on how the formula was rediscovered (section 6).

The underlying mathematics is entirely classical. The fact that exactly one such relation must exist at $N = pq$ follows immediately from $\dim \mathcal{E}_2(\Gamma_0(pq)) = 3$. The explicit $\gcd(pq - 1, q - p)$ parameterization for the integer-lowest-height representative may not appear in this form in the standard references; if it does, we have not located it. The note is offered both as a convenient explicit form and as a worked example of LLM-assisted autonomous rediscovery of a classical structural identity.

2. Closed-form coefficients

2.1. **The five-line derivation.** Identity (1) collapses each conjugate-pair block to a three-parameter affine function of (α, β) alone. Substituting the conjugate pair $(d\alpha,\, \beta/d)$ into eq. (1)—which is permissible because $d\alpha \cdot \beta/d = \alpha\beta = \pi^2$—we obtain

$$T_d(\alpha, \beta) = \frac{d\alpha + \beta/d}{24} - \frac{1}{4} = \frac{d\,\alpha}{24} + \frac{\beta}{24\,d} - \frac{1}{4}. \tag{5}$$

A linear combination $\sum_{d \mid pq} c_d\, T_d$ therefore equals

$$\frac{\alpha}{24} \sum_d d\, c_d + \frac{\beta}{24} \sum_d \frac{c_d}{d} - \frac{1}{4} \sum_d c_d. \tag{6}$$

The combination vanishes identically in (α, β) iff the three coefficient sums all vanish:

$$\sum_d c_d = 0, \qquad \sum_d d\, c_d = 0, \qquad \sum_d \frac{c_d}{d} = 0. \tag{7}$$

With $d \in \{1, p, q, pq\}$ and unknowns (c_1, c_p, c_q, c_{pq}), labelling eq. (7) as E_1, E_2, E_3 respectively, this is a 3×4 rational linear system. Five lines of Gaussian elimination solve it:

(1) $E_2 - E_1$: $\quad (p - 1)c_p + (q - 1)c_q + (pq - 1)c_{pq} = 0.$
(2) $-pq(E_3 - E_1)$: $\quad q(p - 1)c_p + p(q - 1)c_q + (pq - 1)c_{pq} = 0.$
(3) Step 1 − Step 2: $\quad -(p - 1)(q - 1)(c_p + c_q) = 0$, so $c_q = -c_p$.
(4) Substituting into Step 1: $\quad (pq - 1)c_{pq} = (q - p)\, c_p$, i.e. $m\,(q - p) = \delta\,(pq - 1)$ with $c_p =: m$ and $c_{pq} =: \delta$.
(5) Substituting into E_1 with $c_q = -c_p$: $\quad c_1 = -c_{pq} = -\delta$.

The integer-lowest-height representative is then obtained by dividing through by $g := \gcd(pq-1,\, q-p)$, yielding $\delta = (q-p)/g$, $m = (pq-1)/g$, and the coefficient vector $(c_1, c_p, c_q, c_{pq}) = (-\delta,\, m,\, -m,\, \delta)$ as claimed. This completes the proof of theorem 1.1. □

2.2. **Worked example:** $N = 33 = 3 \cdot 11$**.** Here $p = 3$, $q = 11$, so $q - p = 8$, $pq - 1 = 32$, and $g = \gcd(32, 8) = 8$. Therefore $\delta = 8/8 = 1$ and $m = 32/8 = 4$, giving

$$-T_1(\alpha, \beta) + 4\,T_3(\alpha, \beta) - 4\,T_{11}(\alpha, \beta) + T_{33}(\alpha, \beta) = 0. \tag{8}$$

Expanding the blocks explicitly,

$$\alpha\big[-L(\alpha) + 12\,L(3\alpha) - 44\,L(11\alpha) + 33\,L(33\alpha)\big] + \beta\big[-L(\beta) + \tfrac{4}{3}\,L(\beta/3) - \tfrac{4}{11}\,L(\beta/11) + \tfrac{1}{33}\,L(\beta/33)\big] = 0.$$

Numerically, at the canonical cell $\alpha = 1$, $\beta = \pi^2$, the left-hand side evaluates to magnitude $\le 10^{-199}$ at `mpmath` working precision 200 digits, against blocks of magnitude $\sim 10^{-1}$—relative error $\le 10^{-199}$.

3. Connection to classical modular forms

3.1. **The dimension formula.** For squarefree $N = q_1 q_2 \cdots q_t$ with t distinct prime factors, the Eisenstein subspace of weight-2 modular forms on $\Gamma_0(N)$ has dimension

$$\dim \mathcal{E}_2(\Gamma_0(N)) = 2^t - 1. \tag{9}$$

This is a standard fact; see Diamond–Shurman [5, §3.5] or Stein [9, Ch. 5]. For $t = 2$ (the squarefree composite case $N = pq$) this gives $\dim \mathcal{E}_2(\Gamma_0(pq)) = 3$.

The four functions $\{E_2(\tau),\ E_2(p\tau),\ E_2(q\tau),\ E_2(pq\tau)\}$ are weight-2 Eisenstein-style forms on $\Gamma_0(pq)$. Each $E_2(d\tau)$ is a quasi-modular form on $\Gamma_0(d) \supseteq \Gamma_0(pq)$, and the standard old-form construction places it in the Eisenstein subspace of $M_2(\Gamma_0(pq))$ after subtracting the appropriate quasi-modular anomaly; see [5, §5.6]. Because four candidate forms live in a three-dimensional space, they must satisfy at least one linear relation. The argument of section 2.1—which used only eq. (1), not the Eisenstein-subspace framework—shows that the linear-relation space is exactly one-dimensional.

The LMFDB database [10] confirms the dimension count at $N = 33$: the space $M_2(\Gamma_0(33))$ has total dimension 9, cuspidal subspace dimension 6, and Eisenstein subspace dimension 3.

3.2. **The Hardy–Ramanujan–Berndt identity in this light.** Identity (1) is precisely the manifestation, at the level of one conjugate pair of Lambert blocks, of the transformation law (2) for E_2. The anomaly $6\tau/(\pi i)$ in eq. (2) becomes the constant correction $-1/4$ and the linear piece $(\alpha + \beta)/24$ in eq. (1). The fact that this anomaly is *itself a polynomial of low degree in* (α, β)—specifically of degree at most one—is what reduces the four-divisor annihilator equation to only three linear constraints (eq. (7)), and hence to a kernel of dimension $4 - 3 = 1$ in the generic case.

Said differently: the Hardy–Ramanujan–Berndt identity is the dimension reduction $\dim \mathcal{E}_2(\Gamma_0(pq)) = 3$ in disguise, working at the level of Lambert summands rather than modular forms.

4. Numerical verification

We verified eq. (4) across sixteen instances using `mpmath` at working precision 200 digits, evaluating the left-hand side at five cells on the hyperbola $\alpha\beta = \pi^2$ chosen to stress the formula at small α, large α, and the symmetric point $\alpha = \beta = \pi$. The strict acceptance bar is relative error $\le 10^{-100}$, where the relative error is measured against $\max_d |c_d \cdot T_d|$ (the cancellation scale).

Table 1 summarizes seven main instances plus nine adversarial edge cases.

The bar-marginal entries at $N = 323$ (twin primes $17 \cdot 19$) and $N = 391$ (far-apart primes $17 \cdot 23$) reflect numerical cancellation eating ~ 66 and ~ 90 digits respectively in the most extreme cell; both remain comfortably inside the 10^{-100} bar. All other entries achieve relative error $\le 10^{-199}$, i.e. essentially the working-precision floor.

The formula is symmetric under $p \leftrightarrow q$: the substitution flips $\delta \to -\delta$ and the labels $c_p \leftrightarrow c_q$, leaving the relation invariant up to an overall sign. The derivation in section 2.1 nowhere assumed $p > 2$; the small-prime entries $N \in \{6, 10, 14, 22\}$ confirm this.

N	(p,q)	Coefficients (c_1,c_p,c_q,c_{pq})	g, δ, m	Worst rel. err. / 5 cells
33	$(3,11)$	$(-1,4,-4,1)$	$8,1,4$	$\le 10^{-199}$
35	$(5,7)$	$(-1,17,-17,1)$	$2,1,17$	2.76×10^{-200}
51	$(3,17)$	$(-7,25,-25,7)$	$2,7,25$	2.85×10^{-200}
65	$(5,13)$	$(-1,8,-8,1)$	$8,1,8$	4.42×10^{-200}
77	$(7,11)$	$(-1,19,-19,1)$	$4,1,19$	1.27×10^{-200}
87	$(3,29)$	$(-13,43,-43,13)$	$2,13,43$	1.99×10^{-199}
95	$(5,19)$	$(-7,47,-47,7)$	$2,7,47$	4.88×10^{-200}
115	$(5,23)$	$(-3,19,-19,3)$	$6,3,19$	1.07×10^{-199}
143	$(11,13)$	$(-1,71,-71,1)$	$2,1,71$	1.93×10^{-200}
323	$(17,19)$	$(-1,161,-161,1)$	$2,1,161$	6.99×10^{-134}
187	$(11,17)$	$(-1,31,-31,1)$	$6,1,31$	9.60×10^{-200}
391	$(17,23)$	$(-1,65,-65,1)$	$6,1,65$	3.46×10^{-110}
6	$(2,3)$	$(-1,5,-5,1)$	$1,1,5$	4.02×10^{-200}
10	$(2,5)$	$(-1,3,-3,1)$	$3,1,3$	1.91×10^{-200}
14	$(2,7)$	$(-5,13,-13,5)$	$1,5,13$	2.70×10^{-200}
22	$(2,11)$	$(-3,7,-7,3)$	$3,3,7$	1.22×10^{-199}

TABLE 1. Numerical verification of eq. (4) across sixteen $N=pq$ instances at `mpmath` working precision 200 digits. Worst relative error is the maximum, across five test cells on $\alpha\beta=\pi^2$, of $|\mathrm{LHS}|/\max_d|c_d\,T_d|$.

5. Structural extensions

What happens at richer divisor sets? Section 2.1 shows that any divisor set $D=\{d_1,\dots,d_n\}$ on which T_d is defined yields the same three linear constraints (7), whose kernel has dimension $n-\mathrm{rank}(M)$ where M is the $3\times n$ matrix with rows $(1,\dots,1)$, $(d_1,\dots,d_n)$, $(1/d_1,\dots,1/d_n)$. The interesting structural extensions are:

Squarefree $N=pqr$ (three distinct primes). Eight divisors $\{1,p,q,r,pq,pr,qr,pqr\}$, a 3×8 rational system, kernel dimension 5. The closed-form parameterization of theorem 1.1 does *not* extend uniquely; the kernel is non-unique. Numerically verified at $N\in\{30,\,105,\,165\}$. The first basis vector is typically a "sub-divisor annihilator" supported on $\{1,p,q,r\}$ alone; the lifted theorem 1.1 annihilators on the embedded pq-, pr-, qr-sublattices also live inside the kernel.

Prime power $N=p^2$. Three divisors $\{1,p,p^2\}$. The 3×3 constraint matrix is Vandermonde-type in $(1,p,p^2)$ and $(1,1/p,1/p^2)$, of full rank 3, kernel dimension 0. *The family does not extend to prime powers.* Verified at $N\in\{49,121\}$. The construction structurally requires at least two distinct prime factors.

Mixed $N=p^2q$. Six divisors $\{1,p,q,p^2,pq,p^2q\}$, kernel dimension 3. The closed form is non-unique. Verified at $N\in\{18,45\}$. As at pqr, sub-divisor annihilators and lifts of the theorem 1.1 $p\cdot q$ annihilator both populate the kernel.

The clean uniqueness of theorem 1.1 is therefore special to the squarefree composite case $N=pq$.

Remark 5.1. A subtler point: at $N=pqr$ the Eisenstein-subspace dimension formula (9) gives $\dim\mathcal{E}_2(\Gamma_0(pqr))=2^3-1=7$, so the eight forms $\{E_2(d\tau):d\mid pqr\}$ satisfy exactly one linear relation at the level of modular forms—yet our Lambert-block construction yields a kernel of dimension 5, not 1. The discrepancy is because eq. (1) captures only the affine truncation of the full E_2 q-expansion under modular inversion; it loses information about the higher-order terms that modular-form linear-relation theory tracks in full. At $N=pq$ the kernel dimensions coincide ($1=1$); at higher N they generically diverge. This is a useful caveat: the Hardy–Ramanujan–Berndt identity is a *strictly weaker* constraint than full E_2-modularity, and the Lambert-annihilator picture and the Eisenstein-subspace linear-relation picture only agree at $N=pq$.

6. Provenance: autonomous rediscovery by symbolic search

The closed-form formula (4) was rediscovered autonomously by Project C, an LLM-driven symbolic-search engine for Ramanujan-style identities developed during the 2026 *Ramanujan mining* research run [7]. The engine's basis of operators—`mellin_inversion`, `master_theorem`, `modular_transform_with_correction`, and others—was applied iteratively to a small library of seed identities under an `mpmath` cost model with a strict $\leq 10^{-30}$ relative-error acceptance bar. The Level-33 instance was the first the engine surfaced (after several thousand seed-operator compositions); a prompted exploration cycle then enumerated the seven additional instances $N \in \{35, 51, 65, 77, 87, 95, 115\}$ from a hand-tabulated guess of the integer-lowest-height coefficient pattern, yielding the explicit $\gcd(pq-1, q-p)$ form.

Crucially, the engine did *not* use the classical dimension formula (9) or the Eisenstein-subspace framework of section 3. It worked entirely from eq. (1) and elementary linear algebra over the rationals. The structural connection to $\dim \mathcal{E}_2(\Gamma_0(pq)) = 3$ was identified only *after* the formula was produced, by an adversarial verification subagent performing a literature audit. Where helpful, the run also made use of OpenAI's frontier-model interface as a complementary sounding board [8].

A subsequent literature diff [7, diary 58] across six volumes of Berndt's *Ramanujan's Notebooks* and *Lost Notebook* commentary [2, 1], Berndt–Yee [4], Berndt–Dixit [3], plus the LMFDB database, the OEIS, and approximately twelve targeted arXiv and AMS searches, did not locate the explicit $\gcd(pq-1, q-p)$ parameterization in this exact form. The underlying *structure*—a one-dimensional linear-relation space among the four Eisenstein candidates at squarefree composite level—is, however, classical and well-documented.

We therefore offer this note in two registers. First, as a useful explicit closed form for a relation that is *guaranteed to exist* by Eisenstein-subspace dimension counting but is rarely written down in this elementary form, and which evaluates exactly to zero at every (α, β) on $\alpha\beta = \pi^2$. Second, as a worked instance of autonomous rediscovery: an LLM-driven engine, operating with no prior modular-forms machinery and only a small basis of structural operators, rediscovered in explicit form a classical structural identity that has been implicit in the modular-forms literature for decades. We regard the latter as the more interesting contribution. The mathematics is classical; what is novel is the demonstration that a small, structured operator basis plus a strict numerical acceptance gate suffices to surface this structure autonomously.

The full computational verification—numerical code, raw results, and the literature audit—is available at [7] in `analysis/30_level33_lambert_e2_annihilator_verification.py` and `analysis/32_level_pq_fam`

Acknowledgments

The author thanks the developers of `mpmath`, `sympy`, the LMFDB project, and the maintainers of the open-access portions of Berndt's *Ramanujan's Notebooks* series.

References

1. George E. Andrews and Bruce C. Berndt, *Ramanujan's lost notebook, parts i–iv*, Springer, New York, 2005–2013.
2. Bruce C. Berndt, *Ramanujan's notebooks, part ii*, Springer-Verlag, New York, 1989, Chapter 14, Entries 26–27 cover the Hardy–Ramanujan–Berndt Lambert/E_2 identity used throughout this note.
3. Bruce C. Berndt and Atul Dixit, *A transformation formula involving the gamma and Riemann zeta functions in Ramanujan's lost notebook*, The Legacy of Alladi Ramakrishnan in the Mathematical Sciences, Springer, New York, 2010, Preprint at `arXiv:0904.1053`., pp. 199–210.
4. Bruce C. Berndt and Ae Ja Yee, *A page on Eisenstein series in Ramanujan's lost notebook*, RIMS Kôkyuûroku **1274** (2002), 138–153.
5. Fred Diamond and Jerry Shurman, *A first course in modular forms*, Graduate Texts in Mathematics, vol. 228, Springer, New York, 2005, See §3.5, §3.9, §5.6 for the Eisenstein-subspace dimension formula and the old-form construction at composite level.
6. G. H. Hardy and S. Ramanujan, *Asymptotic formulae in combinatory analysis*, Proceedings of the London Mathematical Society (2) **17** (1918), 75–115, Submitted 1916; the base identity used here is equation (1.55).

7. Jouni Helminen, *Project c / `ramanujan-mining`: a private symbolic-search and adversarial-verification research run*, GitHub repository, 2026, https://github.com/jounih/ramanujan-mining; see `diary/55_*.md` (original engine claim), `diary/58_*.md` (adversarial verification and literature audit), `insights/DID_NOT_PAN_OUT.md` §R19, and `analysis/30_*.py`, `analysis/32_*.py` (verification code) at commit `bf5aacf` or later.
8. OpenAI, *ChatGPT (frontier model interface)*, https://chat.openai.com, 2026, Used in May 2026 as a complementary sounding board for prior-art search and structural commentary during the Project C run.
9. William A. Stein, *Modular forms, a computational approach*, Graduate Studies in Mathematics, vol. 79, American Mathematical Society, Providence, RI, 2007, Chapter 5 treats the Eisenstein subspace at squarefree level.
10. The LMFDB Collaboration, *The L-functions and modular forms database*, https://www.lmfdb.org, 2026, Modular form space $M_2(\Gamma_0(33))$: total dimension 9, cuspidal 6, Eisenstein 3 (accessed May 2026).

Independent researcher
Email address: jouni@helminen.co